확률·랜덤프로세스 기말대비 6/14 (D8) · 2차 — 약점 집중 + 중간범위 L01–04 통합

① 오답노트.md 어제 오답부터 재풀이 ② 아래 신규 문제는 '상한'(시간 부족 시 신규만 이월) ③ 채점: origin 풀이 / /solve-problem

떠올릴 개념

• 개념_*.pdf(슬라이드/치트시트)로 핵심 공식을 먼저 최종 점검한 뒤 풀이 시작

문항 목차 (총 12문항)

Problem 18
Let X and Y be two jointly continuous random variables with joint PDF

$$f_{XY}(x,y) = \begin{cases} \frac{1}{4}x^2 + \frac{1}{6}y & -1 \leq x \leq 1, 0 \leq y \leq 2 \\ \\ 0 & \text{otherwise} \end{cases}$$

a. Find the marginal PDFs, $f_X(x)$ and $f_Y(y)$.
b. Find $P(X > 0, Y < 1)$.
c. Find $P(X > 0 \text{ or } Y < 1)$.
d. Find $P(X > 0 | Y < 1)$.
e. Find $P(X + Y > 0)$.

— 풀이 —

5.5.5 X and Y are random variables with the joint PDF

$$f_{X,Y}(x,y) = \begin{cases} 5x^2/2 & \substack{-1 \leq x \leq 1; \\ 0 \leq y \leq x^2} \\ 0 & \text{otherwise.} \end{cases}$$

(a) What is the marginal PDF $f_X(x)$?

(b) What is the marginal PDF $f_Y(y)$?

— 풀이 —

5.7.5 ● X and Y are random variables with $\mathrm{E}[X] = \mathrm{E}[Y] = 0$ and $\mathrm{Var}[X] = 1$, $\mathrm{Var}[Y] = 4$ and correlation coefficient $\rho = 1/2$. Find $\mathrm{Var}[X + Y]$.

— 풀이 —

5.8.5 X and Y are independent random variables with PDFs

$$f_X(x) = \begin{cases} \frac{1}{3}e^{-x/3} & x \geq 0, \\ 0 & \text{otherwise,} \end{cases}$$

$$f_Y(y) = \begin{cases} \frac{1}{2}e^{-y/2} & y \geq 0, \\ 0 & \text{otherwise.} \end{cases}$$

(a) Find the correlation $r_{X,Y}$.

(b) Find the covariance $\mathrm{Cov}[X,Y]$.

— 풀이 —

Problem 38

Let X and Y be jointly normal random variables with parameters $\mu_X = 2$, $\sigma_X^2 = 4$, $\mu_Y = 1$, $\sigma_Y^2 = 9$, and $\rho = -\frac{1}{2}$.

a. Find $E[Y|X = 3]$.
b. Find $Var(Y|X = 2)$.
c. Find $P(X + 2Y \leq 5|X + Y = 3)$.

← previous
next →

— 풀이 —

7.5.6 Random variables X and Y have joint PDF

$$f_{X,Y}(x,y) = \begin{cases} 1/2 & -1 \leq x \leq y \leq 1, \\ 0 & \text{otherwise.} \end{cases}$$

(a) What is $f_Y(y)$?

(b) What is $f_{X|Y}(x|y)$?

(c) What is $\mathrm{E}[X|Y=y]$?

— 풀이 —

1.6.5 In an experiment, A and B are mutually exclusive events with probabilities $\mathrm{P}[A] = 1/4$ and $\mathrm{P}[B] = 1/8$.

(a) Find $\mathrm{P}[A \cap B]$, $\mathrm{P}[A \cup B]$, $\mathrm{P}[A \cap B^c]$, and $\mathrm{P}[A \cup B^c]$.

(b) Are A and B independent?

— 풀이 —

3.5.16◆ In a TV game show, there are three identical-looking suitcases. The first suitcase has 3 dollars, the second has 30 dollars and the third has 300 dollars. You start the game by randomly choosing a suitcase. *Between the two unchosen suitcases*, the game show host opens the suitcase with more money. The host then asks you if you want to keep your suitcase or switch to the other remaining suitcase. After you make your decision, you open your suitcase and keep the D dollars inside. Should you switch suitcases? To answer this question, solve the following subproblems and use the following notation:

- C_i is the event that you first choose the suitcase with i dollars.
- O_i denotes the event that the host opens a suitcase with i dollars.

In addition, you may wish to go back and review the Monty Hall problem in Example 2.4.

(a) Suppose you never switch; you always stick with your original choice. Use a tree diagram to find the PMF $P_D(d)$ and expected value $\mathrm{E}[D]$.

(b) Suppose you always switch. Use a tree diagram to find the PMF $P_D(d)$ and expected value $\mathrm{E}[D]$.

(c) Perhaps your rule for switching should depend on how many dollars are in the suitcase that the host opens? What is the optimal strategy to maximize $\mathrm{E}[D]$? Hint: Consider making a random decision; if the host opens a suitcase with i dollars, let α_i denote the probability that you switch.

— 풀이 —

3.8.1 In an experiment to monitor two packets, the PMF of N, the number of video packets, is

n	0	1	2
$P_N(n)$	0.2	0.7	0.1

Find $E[N]$, $E[N^2]$, $Var[N]$, and σ_N.

— 풀이 —

5.2.1 ● Random variables X and Y have the joint PMF

$$P_{X,Y}(x,y) = \begin{cases} cxy & x = 1,2,4; \quad y = 1,3, \\ 0 & \text{otherwise.} \end{cases}$$

(a) What is the value of the constant c?

(b) What is $\mathrm{P}[Y < X]$?

(c) What is $\mathrm{P}[Y > X]$?

(d) What is $\mathrm{P}[Y = X]$?

(e) What is $\mathrm{P}[Y = 3]$?

— 풀이 —

5.2.5 In Figure 5.2, the axes of the figures are labeled X and Y because the figures depict possible values of the random variables X and Y. However, the figure at the end of Example 5.3 depicts $P_{X,Y}(x, y)$ on axes labeled with lowercase x and y. Should those axes be labeled with the uppercase X and Y? Hint: Reasonable arguments can be made for both views.

— 풀이 —

5.3.1 Given the random variables X and Y in Problem 5.2.1, find

(a) The marginal PMFs $P_X(x)$ and $P_Y(y)$,

(b) The expected values $E[X]$ and $E[Y]$,

(c) The standard deviations σ_X and σ_Y.

— 풀이 —